03 Regression

Markéta Tranberg, mlap@viauc.dk

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ML problems

- Supervised learning
  - Regression
  - Classification
- Unsupervised learning
  - Clustering
  - Dimensionality reduction

Linear algebra

- EF, RREF
- $A\bar{x} = \bar{b}$
- Matrix multiplication $A \times B : (m \times n)(k \times l)$
- Matrix-vector multiplication
- Matrix transpose
- Matrix inverse

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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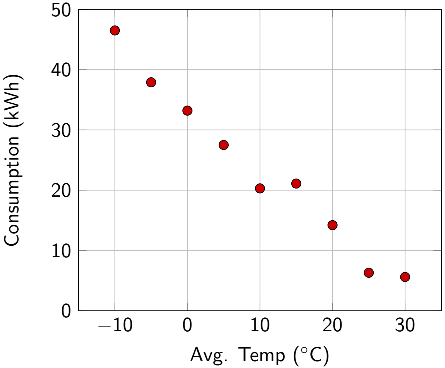
Finding the best line

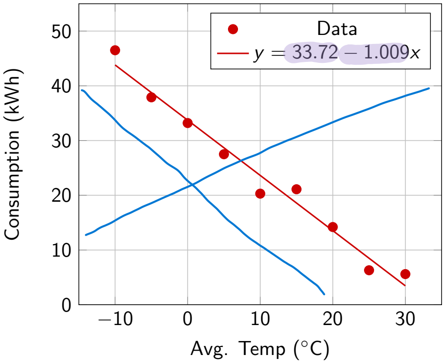
feature — response variable

Avg. Temp (°C)	Daily kWh
-10	46.5
-5	37.9
0	33.2
5	27.5
10	20.3
15	21.1
20	14.2
25	6.3
30	5.6



Regression

x_1	y
-10	46.5
-5	37.9
0	33.2
5	27.5
10	20.3
15	21.1
20	14.2
25	6.3
30	5.6



Regression

$\bar{y}$: (observed) vector

$\hat{y}$: prediction

Given that

$$\hat{y} = \theta_0 + \theta_1 x_1$$

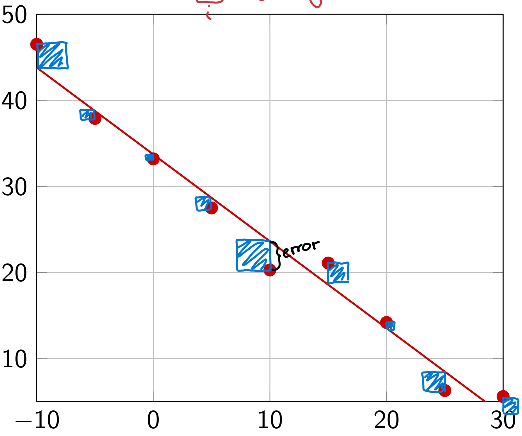
find the "best" values of θ_0 and θ_1

Training the model = finding θ_0 and θ_1
Previous example: $\theta_0 = 33.72, \theta_1 = -1.009$

Least squares problem

Minimization of error (SSE) = sum of square errors

$\sum_i (y^{(i)} - \hat{y}^{(i)})^2$



best line ≈ least SSE

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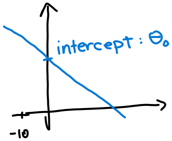
Matrices and vector for regression

$\bar{\Theta}^T \cdot \bar{x}$

theta

$$\hat{y} = \theta_0 + \theta_1 \underline{x_1} + \theta_2 \underline{x_2} + \ldots = \bar{\theta} \cdot \underline{\bar{x}}$$

- $\hat{y}$: predicted value
- x_i: features
- θ_i: model parameters
- $\bar{\theta}$: parameter vector
- $\bar{x}$: feature vector



$x = 0$
$\hat{y} = \Theta_0$

Linear model

$$\hat{y} = \theta_0 \cdot 1 + \theta_1 x_1 + \Theta_1 x_1^2$$

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \dots & \dots \\ 1 & x_n \end{bmatrix} \begin{matrix} x_1^2 \\ \vdots \\ \\ \\ \end{matrix}, \hat{\theta} = \begin{bmatrix} \hat{\theta}_0 \\ \hat{\theta}_1 \end{bmatrix}, \bar{y} = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix}$$

- X: design matrix
- n: sample size
- $\hat{\theta}$: parameter vector
- $\bar{y}$: observation vector

The design matrix has a ~~row~~ of 1s to be able to find the intercept ($\hat{\theta}_0$).
column

Simple linear regression

Simple linear regression model: find estimates of θ_0 and θ_1 that minimizes the **error** (sum of squared errors = SSE), i.e. find $\hat{\theta}_0$ and $\hat{\theta}_1$

Using least squares:

1. X^TX $(n \times m) \times (m \times n) = (n \times n)$
2. $X^T\bar{y}$ $(2 \times n) \times (n \times 1) = 2 \times 1$
3. Solve $\left[\, X^TX \underset{\text{concat}}{,} X^T\bar{y} \,\right] \overset{\text{RREF}}{\approx} \left[\begin{array}{cc} \underbrace{\begin{matrix} 1 & 0 \\ 0 & 1 \end{matrix}}_{I_2} & \underbrace{\begin{matrix} \hat{\theta}_0 \\ \hat{\theta}_1 \end{matrix}}_{\text{parameters}} \end{array}\right]$
4. State $\hat{\theta}_0, \hat{\theta}_1$

Normal equation:

$$\hat{\theta} = (X^TX)^{-1}X^T\bar{y}$$

Example

child systolic pressure

Fit a linear model of the form $\theta_0 + \theta_1 \cdot lnW = P$

W	44	61	81	113	131
P	91	98	103	110	112

$$X = \begin{bmatrix} 1 & \ln(44) \\ 1 & \ln(61) \\ 1 & \vdots \\ 1 & \vdots \\ 1 & \ln(131) \end{bmatrix}$$

desin matrix

$$\bar{P} = \begin{bmatrix} 91 \\ 98 \\ \vdots \\ 112 \end{bmatrix}$$

observation v.

Example

1. $X^TX = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ \ln(44) & \cdots \end{bmatrix} \begin{bmatrix} 1 & \ln(44) \\ 1 & \vdots \\ 1 & \vdots \\ 1 & \\ 1 & \end{bmatrix} = 2 \times 2$

2. $X^T\bar{P} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ \ln(44) & \cdots \end{bmatrix} \begin{bmatrix} 91 \\ \vdots \\ \vdots \end{bmatrix} = 2 \times 1$

3. $\left[\; X^TX \quad X^T\bar{P} \;\right] \overset{RREF}{=} \begin{bmatrix} 1 & 0 & 17.92 \\ 0 & 1 & 19.39 \end{bmatrix}$ $\quad \hat{\theta}_0 = 17.92\ldots$ $\quad \hat{\theta}_1 = 19.39\ldots$

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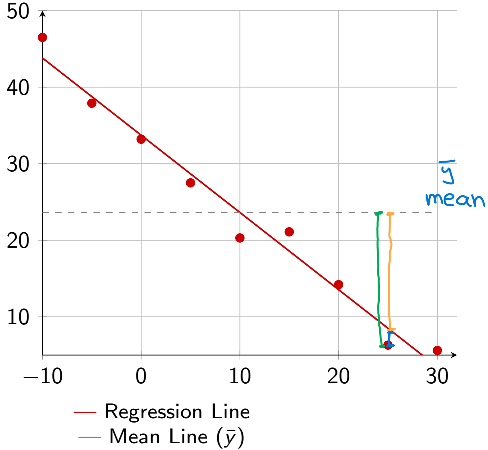
Performance metrics formulas



- $SST = \sum_i (y^{(i)} - \bar{y})^2$
- $SSE = \sum_i (y^{(i)} - \hat{y}^{(i)})^2$
- $SSR = \sum_i (\hat{y}^{(i)} - \bar{y})^2$

Performance metrics:

- $r^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$
- $MSE = \frac{1}{n} \sum_{i=1}^{n} (y^{(i)} - \hat{y}^{(i)})^2$
- $RMSE = \sqrt{MSE}$

Performance metrics explained

- **SST** = total sum of squares
- **SSE** = sum of square errors (unexplained) error
- **SSR** = regression sum of squares (explained)
- **MSE** = mean squared error (mean SSE)
- **RMSE** = root mean squared error (in original units)
- **R2** = coefficient of determination (how much variance the model can explain)

aim $\longrightarrow$ $R^2 = 1$ perfect fit (overfitting)

$R^2 = 0$ predict mean

$R^2 < 0$ worse than $\bar{y}$

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Polynomial regression

$$\hat{y} = \theta_0 + \theta_1 \underbrace{x_1} + \theta_2 \underbrace{x_2} + \theta_3 \underbrace{x_3}$$

features

$x_2 = x_1^2$

$x_3 = x_1^3$

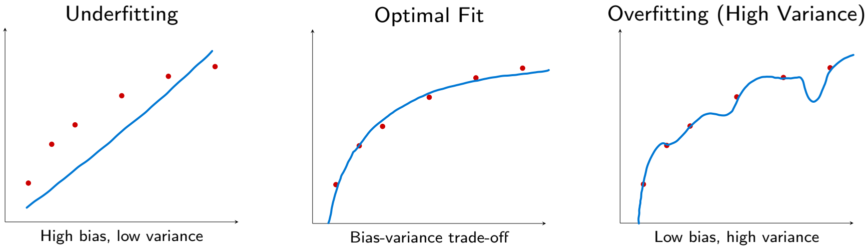
not always a good idea

→ depends on data

Design matrix:

$$X = \begin{bmatrix} 1 & x_{11} & x_{11}^2 & x_{11}^3 \\ 1 & x_{12} & x_{12}^2 & x_{12}^3 \\ 1 & x_{13} & x_{13}^2 & x_{13}^3 \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Overfitting and Underfitting



- **Bias:** inability to learn from data
- **Variance:** reliance on data

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Purpose of regularization

Regularizers avoid overfitting.
Regularization penalizes large coefficients θ (complexity).

Loss function:

$$L = MSE + \alpha R(\theta)$$

- MSE: mean square error
- α: regularization parameter
- $R(\theta)$: penalty function of θ

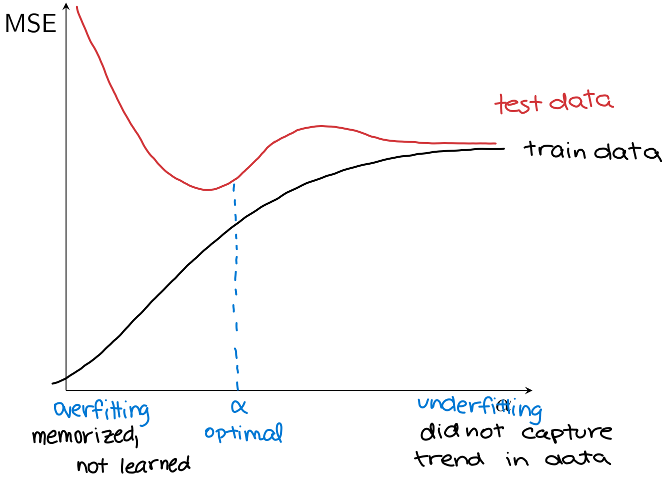
Penalty functions

- **Ridge**
  - L2 regularization ~ Euclidean
  - $R(\theta) = \sum \theta_i^2$
  - Drives coefficients to small values

- **Lasso**
  - L1 regularization ~ Manhattan
  - $R(\theta) = \sum |\theta_i|$
  - Drives some coefficients to zero (feature selection)

- **Elastic net**
  - Combination of L1 and L2
  - Penalty: $\beta \cdot$ Lasso $+ (1 - \beta) \cdot$ Ridge

Optimal regularization parameter



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Code example

Github link: `https://github.com/304125/MAL1_S26/tree/master/03_Regression`